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COURSE INFORMATION

School/Faculty:	Faculty of Computing		
Program name:	BSc in Computer Science (Graphic and Multimedia Software) with honour		
Course code:	SECV2113	**Academic Session/Semester:**	20252026/2
Course name:	Human−Computer Interaction	**Pre/co requisite** (*course name and code, if applicable*):	
Credit hours:	3		

Course synopsis	This is a foundational course in Human−Computer Interaction (HCI) that introduces students to core theories, design processes, and applied practices in user experience (UX) design. It emphasizes both conceptual understanding and practical application. The course explores an evaluation-centered UX lifecycle, emphasizing usability, usefulness, and emotional impact as integral components of user experience. Students will study and apply methods such as contextual inquiry, requirements analysis, design-informing models, design thinking, ideation, sketching, conceptual design, and formative evaluation. The lifecycle is presented as a flexible template adaptable to different development contexts, ensuring relevance to varied project constraints. The course is delivered through a combination of lectures, case studies, and project-driven activities that emphasize hands-on design and iterative evaluation. By the end of the course, students will be able to analyze user needs, design user-centered solutions, and evaluate interactive systems using HCI theories and UX methods. They will also develop the ability to apply iterative design processes to create meaningful, usable, and impactful digital experiences.			
Course coordinator (*If applicable*)	Nur Zuraifah Syazrah bt. Othman. Dr.			
Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail** *@utm.my*
	Nur Zuraifah Syazrah bt. Othman. Dr.	N28 206-04	07-5538827	zuraifah
	Nor Anita Fairos bt. Ismail	N28 206-11	07-5538827	noranita

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	* CLO	PLO ** (MQF Cluster Code)	*** Taxonomies and **** Graduate Attributes	T&L methods	*****Assessment methods
CLO1	Apply the concept of human−computer interaction and the application of human factors as a multi-level process of communication through UX design and evaluation of interactive systems.	PLO1 (C1)	C2 SC1	Lecture, Research-led teaching	Asg., F
CLO2	Analyse the theoretical knowledge to issues that arise in the design of interactive systems.	PLO2 (C2)	C3 SC2	Lecture, Project-based learning	PR, F
CLO3	Build a prototype development project according to principles and guidelines of human−computer interaction.	PLO3 (C3A)	P3 SC3	Project-based learning	PR
CLO4	Ability to act responsibly across the project life cycle by making ethical decisions that prioritise community, environment, and societal well-being.	PLO11 (C5)	A3 A(CC3)	Project-based learning	PA

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.
* Up to 5 CLO
Refer *** Taxonomies of Learning and **** UTM's Graduate Attributes for UG and Generic Skills for PG, where applicable for measurement of outcomes achievement
***** Asg – Assignment; PR – Project; PA - Peer Assessment; F – Final Exam etc.

** **MQF Cluster Code**
C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C**= Communication Skills, **C3D** = Digital Skills, **C3E** = Numeracy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Enterpreneurial Skills, **C5** = Ethics & Professionalism

Details on Innovative T&L practices:

No.	Type	Implementation
1	Project-based learning	Conducted through building a prototype. Tasks are given in sequential steps throughout the semester. Students in a group are asked to build a prototype to solve a challenge faced by users that require HCI and UX knowledge in order to deliver good apps or systems. The report must comply with the project specifications and be given in the form of a written report.
2	Research-led teaching	Research-based approach: Students collect data of various applications, analyse the usability of the applications using evaluation method, interpret the findings and write the report.

Transferable skills (*generic skills learning in course of study which can be useful and utilised in other settings*) :

Able to act responsibly across the project life cycle by making ethical decisions that prioritize community, environment, and societal well-being.

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO*	Learning and Teaching Activities										TOTAL SLT
			Face-to-Face (F2F)								Non F2F Independent Learning		
			Physical				Online (Synchronous)				Online (Asynchronous)	Others	
			L	T	P	O	L	T	P	O			

Week 1	**Introductory to Course** **01 Introduction to Human Computer Interaction** - Good and poor design - Interaction design in comparison to human-computer interaction - User experience, accessibility and inclusiveness - Usability - General design principles	CLO1					2		1			2	5
Week 2	**02 The Process of Interaction Design** - Involving users in development - Principles of user-centred approach - Four basic activities in interaction design - AgileUX - Practical issues *Assignment (Individual)	CLO1					2		1		1	2	6
Week 3	**03 Conceptualising Interaction Design** - Understanding the problem space - Conceptualising the design: # Conceptual models # Metaphors and analogies # Interaction types * Project Part 1	CLO1	2		1							2	5
Week 4	**Section A: Human Behaviour** **04 Users: Cognition Aspects, Social and Emotional Interactions** - Cognition aspects - Social interaction	CLO1	2		1						1	2	6
Week 5	**04 Users: Cognition Aspects, Emotional and Social Interactions (Cont.)** - Emotional interactions * Peer Assessment for Project Part 1	CLO1	2		1							2	5
Week 6	**Section B: Design** **05 Establishing Requirements** - Different kinds of requirements: persona, goals, environments - Data gathering for requirements: interviews, focus groups, questionnaires, ethnography, diaries, contextual inquiry, studying documentation, researching similar products - Data gathering guidelines - Data analysis, interpretation and presentation * Project Part 2	CLO2	2		1						1	2	6
Week 7	**05 Establishing Requirements (Cont.)** - Task description: scenarios, use case - Task analysis: HTA	CLO2 CLO4	2		1							2	5
Week 8	**SEM BREAK**												
Week 9	**06 Trends in Interfaces** - Interfaces types * Peer Assessment for Project Part 2	CLO2	2		1						1	2	6
Week 10	**Section C: Implementation** **07 Design, Prototyping and Construction** - Prototyping and construction - Conceptual design - Concrete design - Scenarios in design - Prototypes in design (storyboard, card-based) - Construction: physical computing, SDK * Project Part 3	CLO3 CLO4	2		1							2	5
Week 11	**07 Design, Prototyping and Construction (Cont.)** - Visual Design Principle - User Journey map	CLO3	2		1						1	2	6
Week 12	**Section D: Evaluation** **08 Introducing Evaluation** - Why, what, where and when of evaluation - Evaluation approaches and methods - Inspection (Heuristic evaluation, walkthrough) - Analytics * Project Part 4a (Build prototype v1) * Prototype v1 Evaluation (Group assignment – same group members as project) * Peer Assessment for Project Part 3	CLO3	2		1							2	5

Week 13	**08 Other Evaluations Types** - Evaluation case studies - How to: usability test, experiment, field studies * Project Part 4b (Prototype v2 Evaluation – test with users)	CLO3 CLO4	2		1						1	2	6
Week 14	**09 Data Analysis, Interpretation and Presentation** - Simple quantitative analysis - Simple qualitative analysis	CLO2	2		1							2	5
Week 15	**09 Data Analysis, Interpretation and Presentation (Cont.)** - Tools to support data analysis * Project Presentation * Peer Assessment for Project Part 4	CLO4	2		1						1	2	6
												SUB-TOTAL SLT :	**77**

	Continuous Assessment	**%**	**Face-to-Face (F2F)**		**Non F2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online** (Synchronous)	**Online** (Asynchronous)	**Others**	
1	Assignment (Group)	5				2	2
2	Project: Part 1	5				3	3
3	Project: Part 2	10				5	5
4	Project: Part 3	10				5	5
5	Project: Part 4	20				10	10
6	Assignment (Individual)	15				5	5
7	Peer Assessments	5				1	1
						SUB-TOTAL SLT :	**31**

	Final Assessment	**%**	**Face-to-Face (F2F)**		**Non F2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online** (Synchronous)	**Online** (Asynchronous)	**Others**	
1	Final Exam	30	3			9	12
							0
							0
							0
						SUB-TOTAL SLT :	**12**
						SLT for Assessment :	**43**
						GRAND TOTAL SLT :	**120**

A	% SLT for F2F Physical Component :	32.50
B	% SLT for Online & Independent Learning Component :	67.50
C	% SLT for Online Component :	10.83
D	% SLT for All Practical Component :	11.67
D1	% SLT for F2F Physical Practical Component :	10.00
D2	% SLT for F2F Online Practical Component :	1.67
Please tick (✓) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (*e.g., software, nursery, computer lab, simulation room etc*)

N/A

References (*include required and further readings, and should be the most current*)

Main references:
Rogers, Y., Sharp, H., Preece, J., (2023) "Interaction Design: Beyond Human-Computer Interaction", 6th Edition, New York: J. Wiley & Sons.

Additional references:
Baxter, Courage & Caine (2015), "Understanding Your Users"
MacKenzie, I.S. (2013), "Human-Computer Interaction: An Empirical Research Perspective", Morgan Kauffman.
Dix, A. et. al (2004), "Human-Computer Interaction", 3rd Edition, Prentice Hall.

Online:
http://elearning.utm.my
Norman Group website https://www.nngroup.com/
Usability.gov website https://www.usability.gov
Interaction Design Foundation https://www.interaction-design.org

Academic honesty and plagiarism:

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES)
Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (*if applicable*)

All students must follow the passing requirements of a computing course: obtain at least the **minimum passing marks** as specified by UTM regulations (**40%** or a grade of **D+**).

Disclaimer:

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ELT = (Theory + Industrial Guidance + Assessment) x 50%
Total of credit for LI/Practical = ELT/40 Notional Hours

Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

Prepared by :		**Certified by:**	
Name :	Nur Zuraifah Syazrah bt. Othman. Dr.	**Name :**	Farhan bin Mohamed, PM Dr
Signature :		**Signature :**	
Date :	11 Mac 2026	**Date :**	